

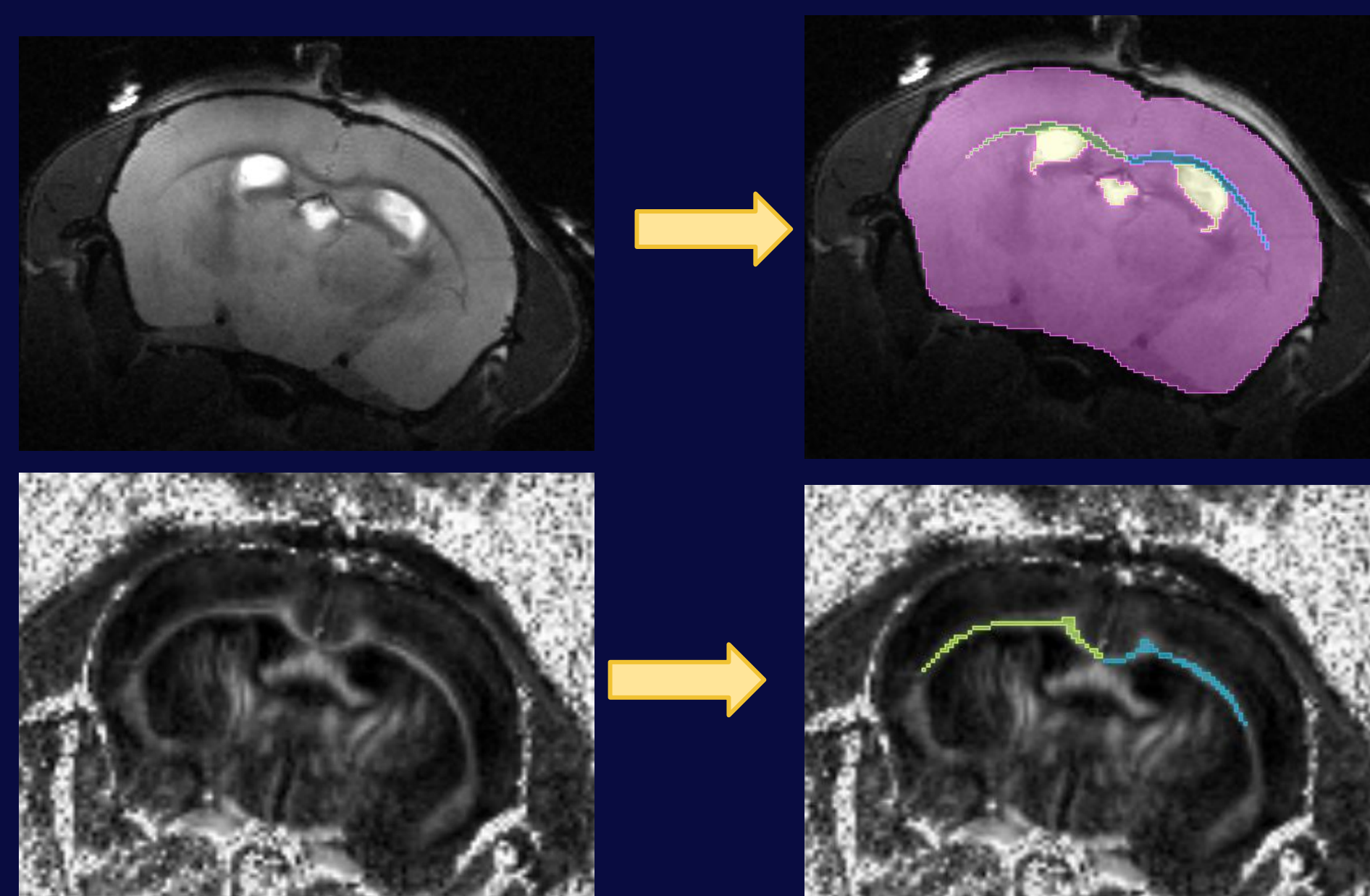
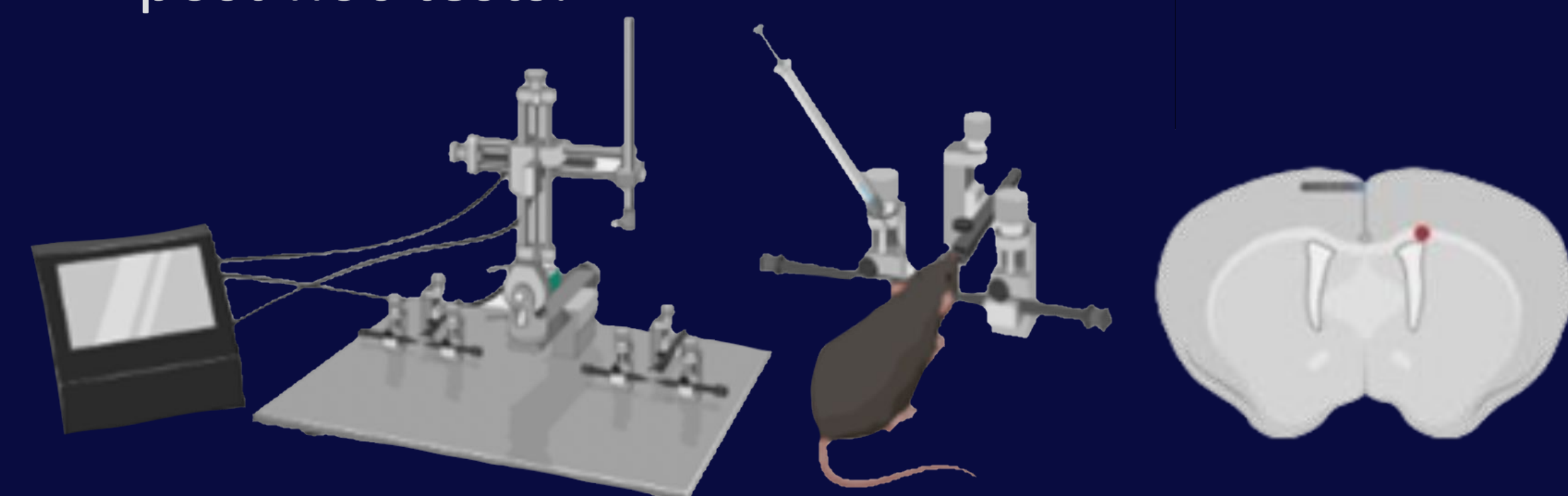
Background

- White matter injury (WMI) significantly contributes to neurological disorders like stroke and age-related cognitive decline.
- CX-4945, a drug that inhibits Casein Kinase 2 (CK2), protects white matter.
- The drug promotes functional recovery in patients of all ages, suggesting a broad therapeutic potential.

We hypothesize that pre-ischemic inhibition of CK2 using CX-4945 can preserve white matter structures and improve recovery outcomes in the selective focal WMI stroke model.

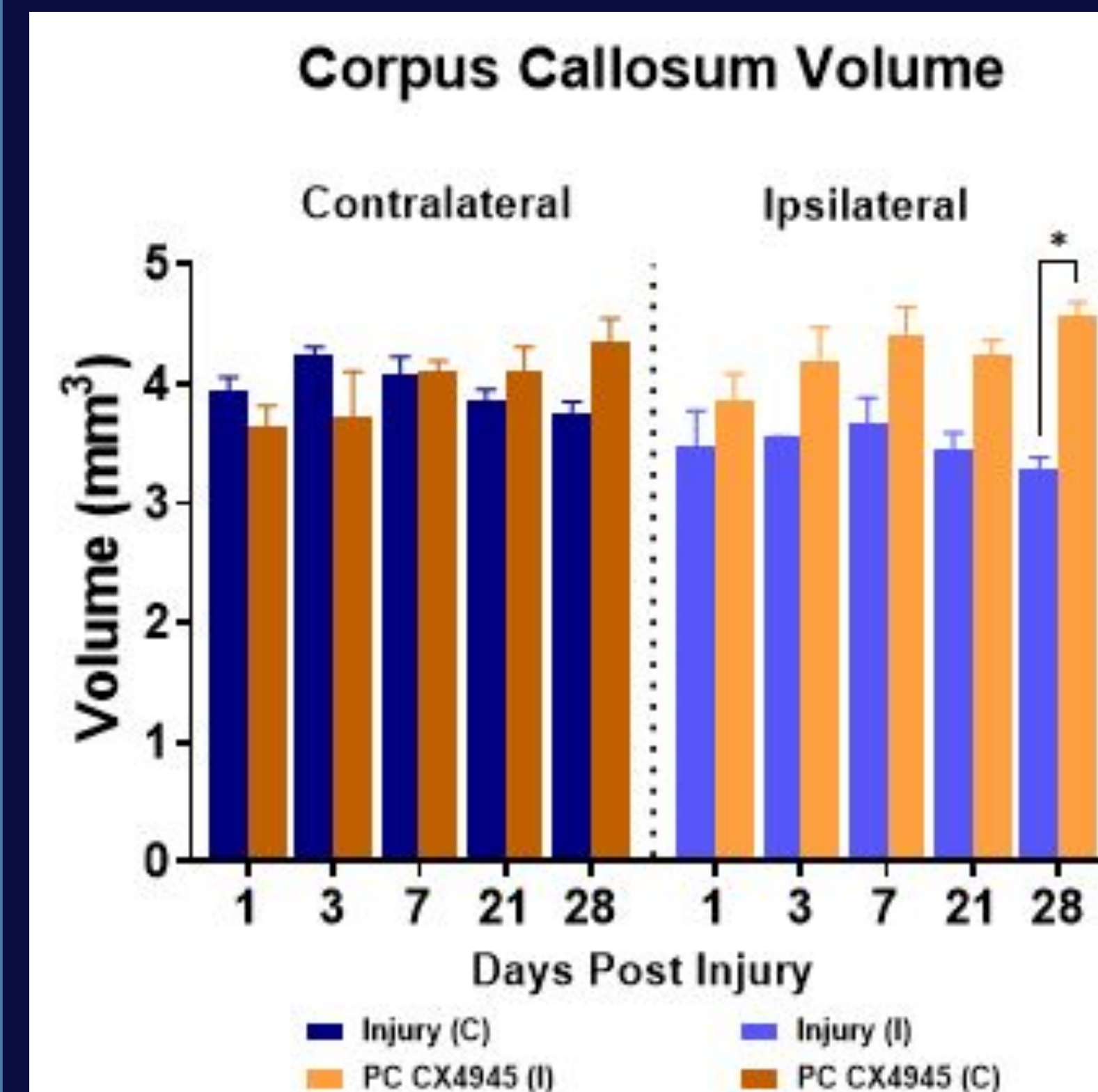
Methods

- Three stereotaxic injections each of 200 nL of L-NIO (130μM) injected at identified coordinates.
- CX-4945 (75 mg/kg) or vehicle twice/day for 5 days orally.
- Longitudinal T2 and DTI MRI scan acquired on days 1,3,7,21 and 28 post-injury.
- 3D Slicer Segmentations of Corpus Callosum(CC)
- Two-way ANOVA test, followed by Tukey's post-hoc tests.

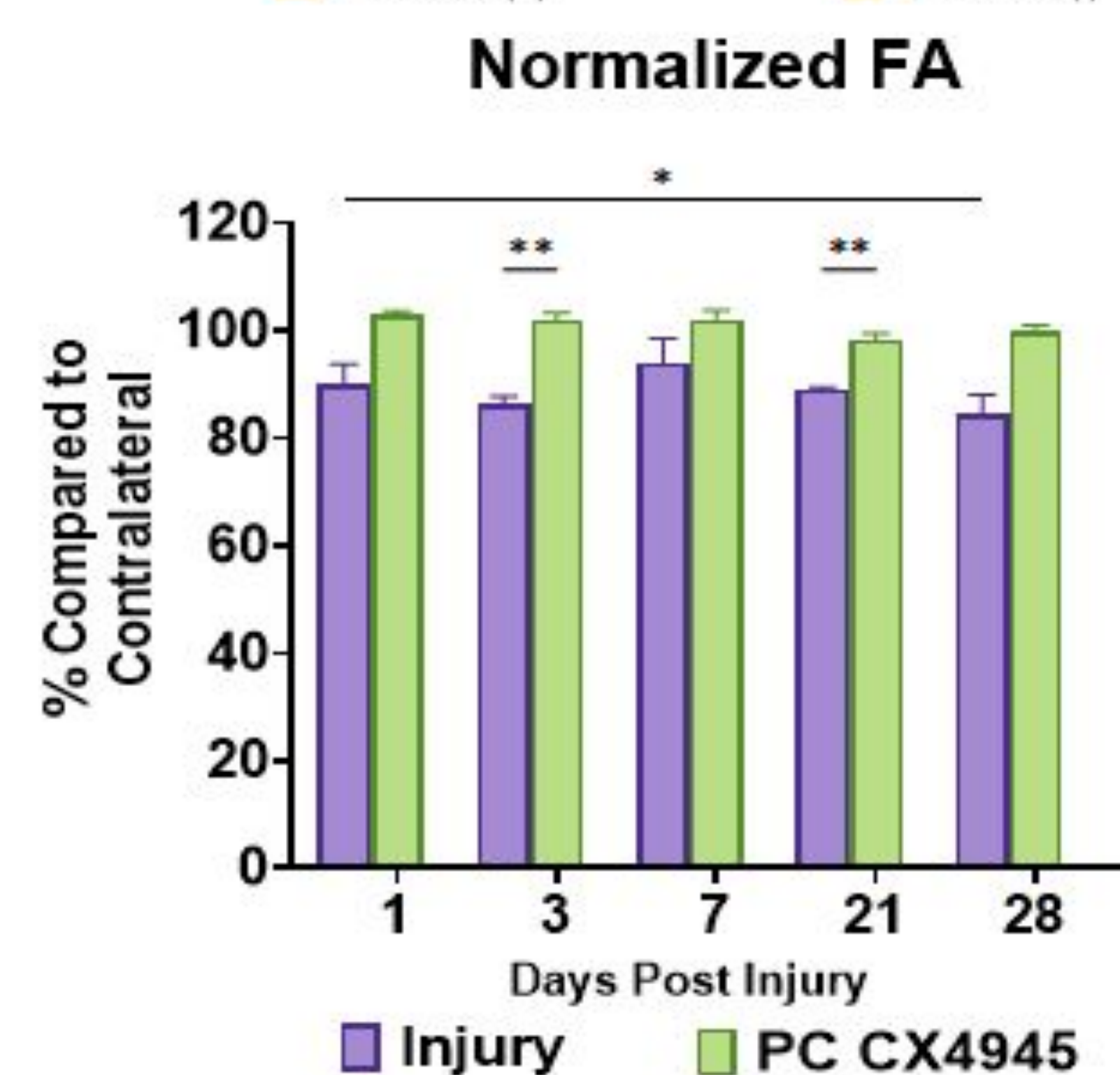
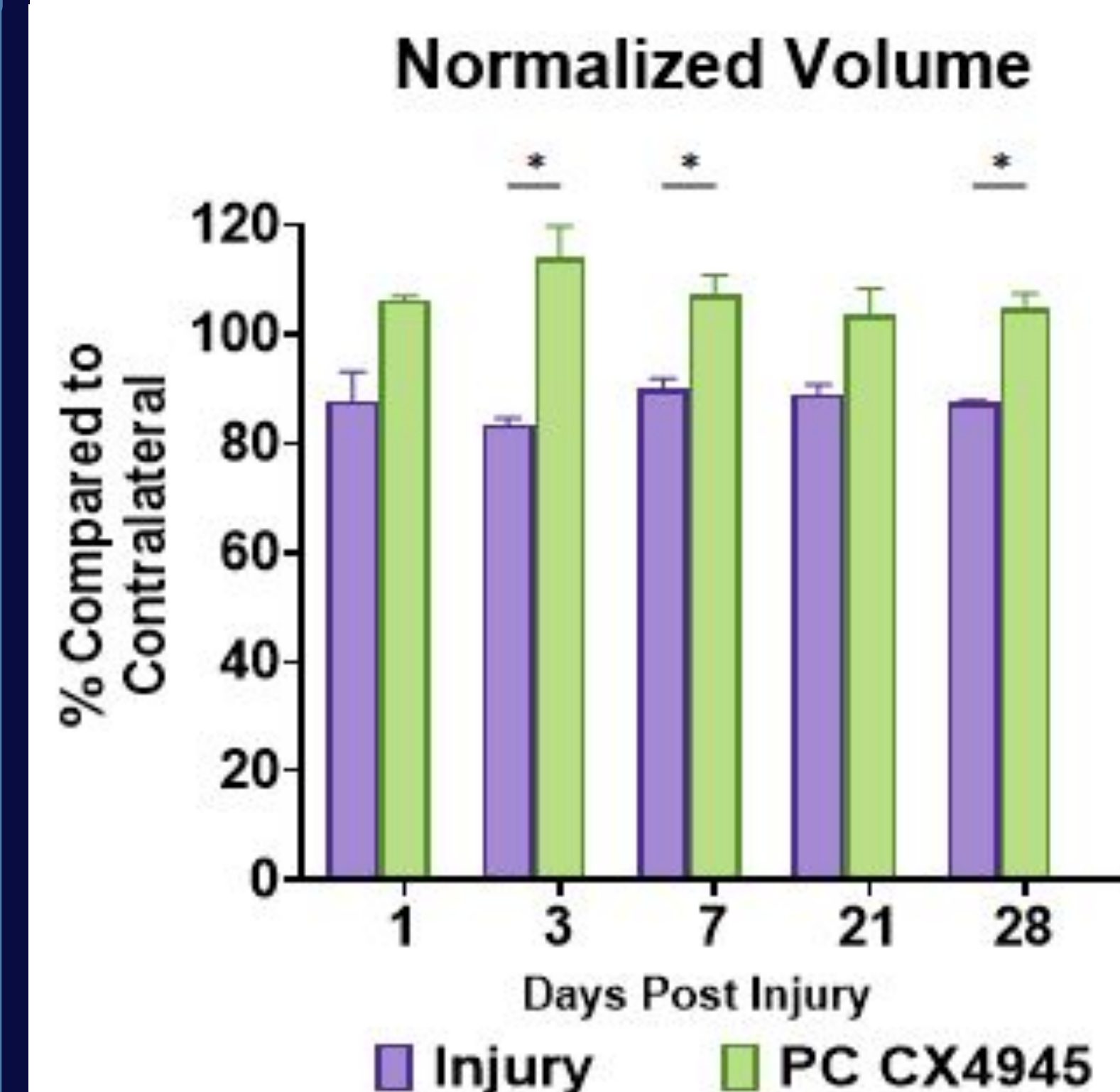
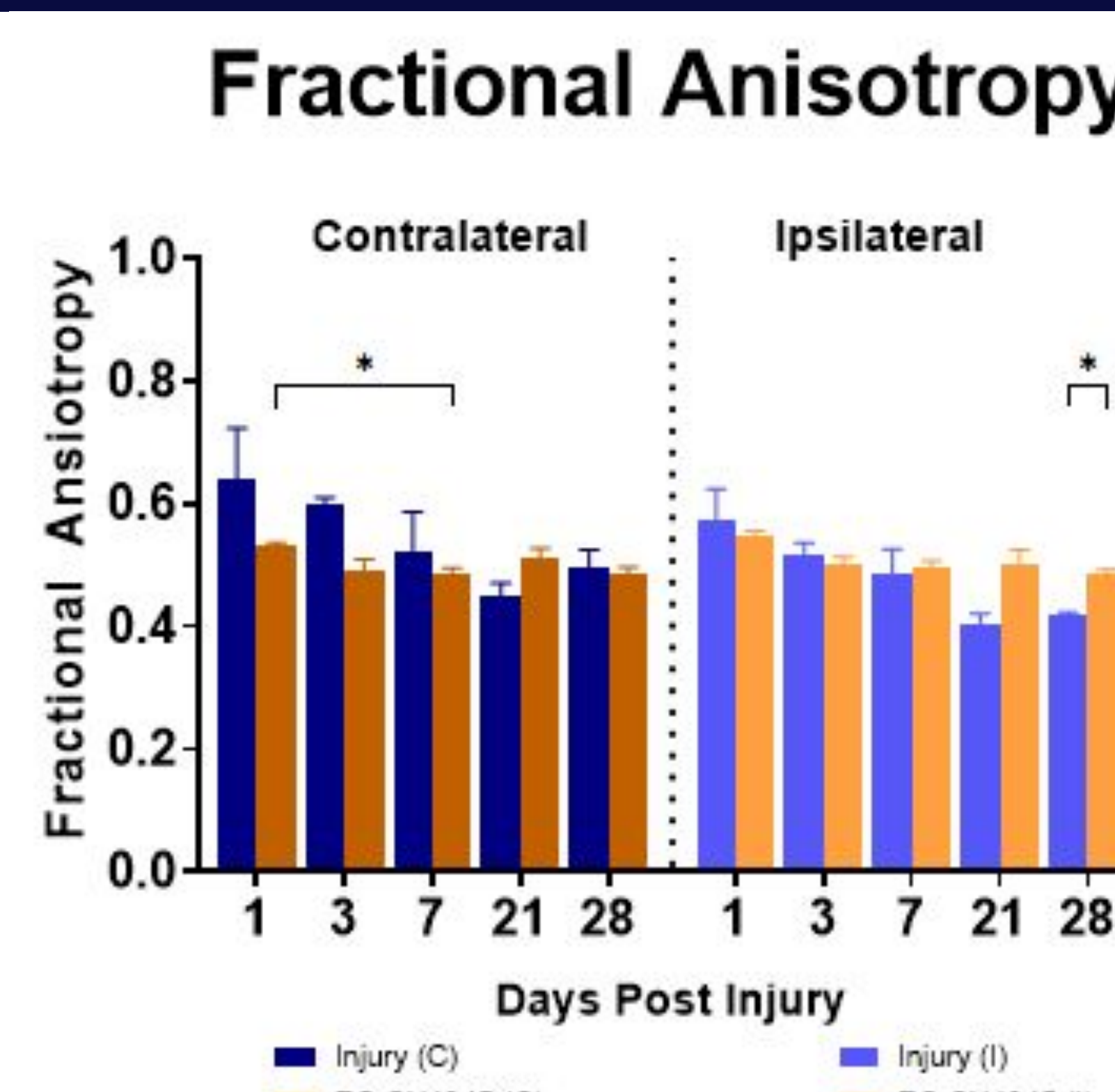


Results: Volume and Intensity

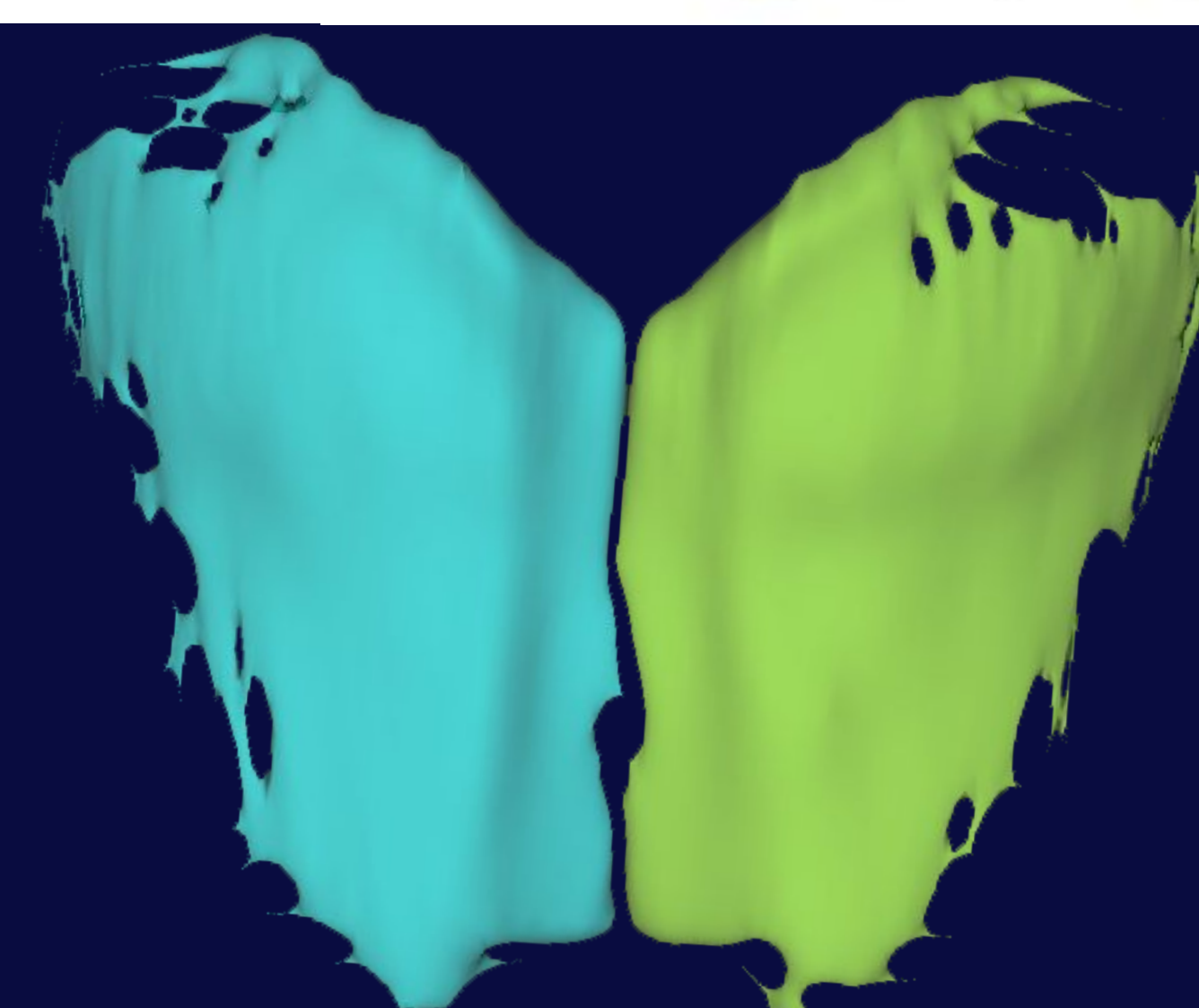
CX-4945 Preconditioning Preserves White Matter Volume



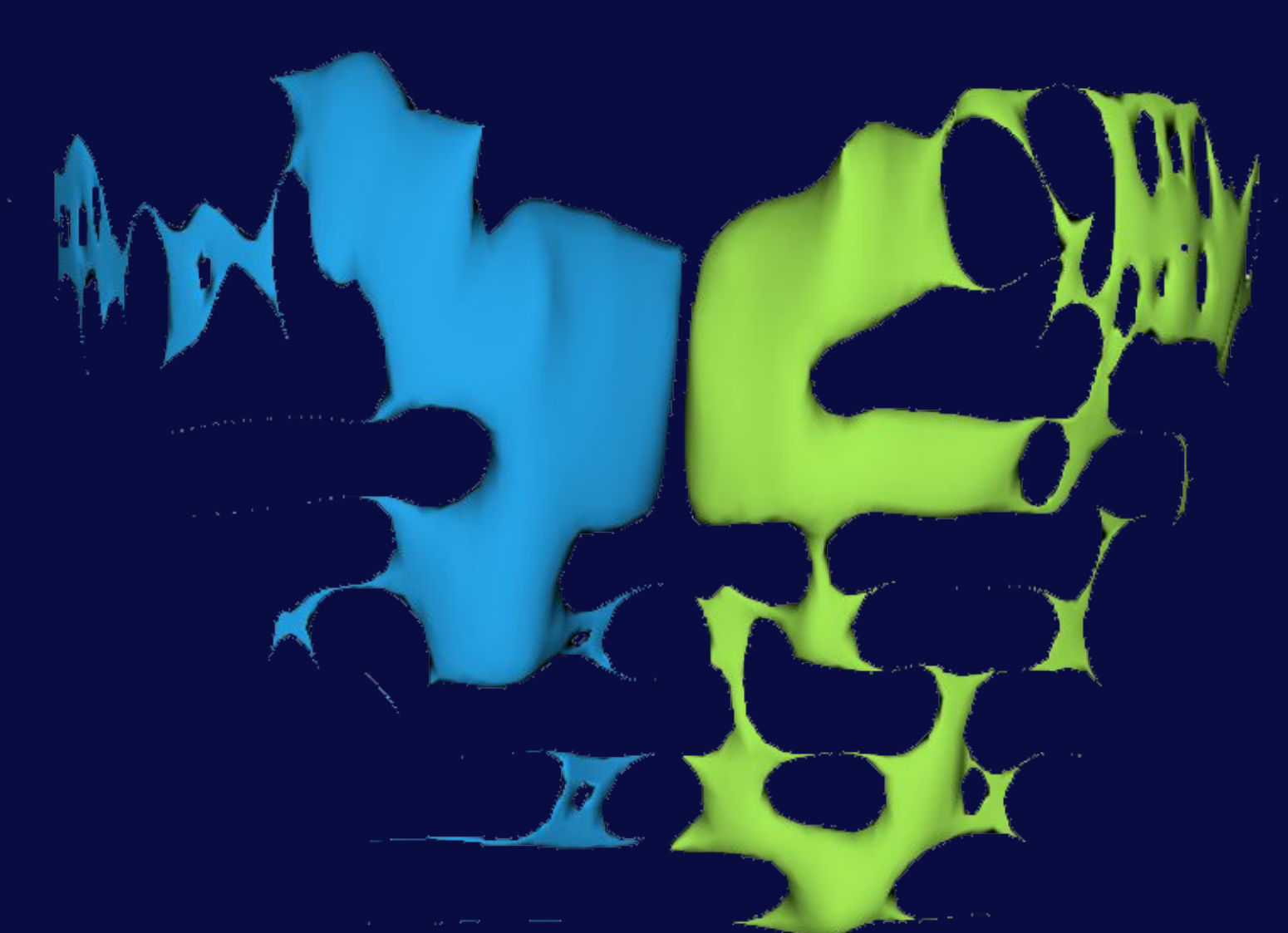
CX-4945 Preconditioning Stabilizes Water Molecule Directionality



Top view of the preconditioned 3D structural CC volume

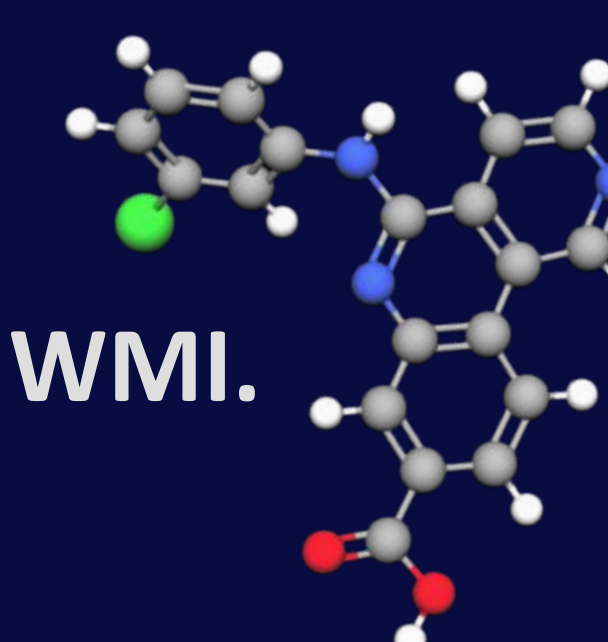


Top view of the injury 3D structural CC volume



Conclusion

- Pre-ischemic CK2 inhibition with CX-4945 preserves WM integrity after selective focal WMI.
- Preconditioning maintains total volume and stabilizes FA values in the ipsilateral CC.
- CX-4945 can be repurposed for stroke therapies.



Future Direction

This study furthers methods to identify therapeutic approaches to alleviate stroke damage and improve the recovery and quality of life for stroke patients.

Acknowledgements

Baltan Lab, Partnership for Scientific Inquiry.